

WatchWise 2.0: Fairness-Aware Group Movie Recommendation with Diffusion Candidates and RL Reranking

Team Number 17

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Abstract. WATCHWISE addresses a common failure mode of movie recommenders: a family or friend group must choose one film together, but standard systems optimize for a single profile or a majority average. We formulate movie night as a fairness-aware group recommendation problem: produce a short slate that preserves overall relevance while protecting the least-satisfied member. The system trains matrix-factorization embeddings on MovieLens 25M, forms reproducible synthetic groups from real users, generates candidates using both nearest-neighbor retrieval and a conditional diffusion model over movie embeddings, and reranks with either a greedy fairness selector or a REINFORCE slate policy. Evaluation is non-circular because held-out user ratings are hidden before training and later used for NDCG@5 and Hit@5. On 101 divergent-taste phase2 test groups, diffusion plus RL achieves the strongest held-out quality (NDCG@5 0.133, Hit@5 0.391) while maintaining worst-member satisfaction above 0.997. A React/FastAPI demo is deployed at <http://34.14.210.127/>.

1 Problem and contribution

Most movie recommenders answer a single-user question: which title should this account watch next? A group decision is different. A family may contain a thriller fan, a comedy fan, a child with age restrictions, and someone with limited time. Simple aggregation can hide minority dissatisfaction: an average-score recommender may select a film that looks good in aggregate while leaving one member with no acceptable option. WATCHWISE instead asks: *what small slate of three to five movies gives the group a good decision while making sure the worst-off member is not stranded?*

The project is designed as a controlled research pipeline and a runnable demo. Its main contributions are:

1. A group recommendation formulation grounded in real MovieLens ratings rather than hand-authored preferences.
2. A controlled candidate-generation comparison between traditional nearest-neighbor (NN) retrieval and conditional denoising diffusion over movie embeddings.
3. A fairness-aware slate objective that combines mean relevance, worst-member coverage, diversity, and disagreement control.
4. Two rerankers over the same candidate pools: a greedy fairness selector and a REINFORCE slate-builder.
5. A non-circular evaluation protocol using ratings hidden before training, plus a deployed React/FastAPI interface. Additional implementation and reproducibility details are in Appendix A.

The key scientific question is deliberately falsifiable: can diffusion-generated candidates improve group compromise quality over NN retrieval when both are judged by the same reranker and the same held-out ground truth? The phase2 answer is yes for held-out NDCG and Hit rate, with the important caveat that the proxy fairness metric is already near its ceiling on MovieLens 25M.

2 Method

WATCHWISE follows a two-stage recommender architecture: generate a candidate pool, then rerank it into a short slate. First, biased matrix factorization learns user and movie embeddings from MovieLens training ratings:

$$\hat{r}_{u,i} = \mu + b_u + b_i + U_u^\top M_i. \quad (1)$$

Synthetic groups are formed from real users as random, similar-taste, or divergent-taste groups. A group embedding is the aggregate of its member embeddings. The system then constructs candidates using either standard NN retrieval around the group embedding or a conditional diffusion model trained on pairs of user-taste vectors and liked movie embeddings. The diffusion model samples embeddings conditioned on the group vector and maps them back to nearest real movies in the catalog. This gives diffusion a clear role: it can propose movies near the learned compromise manifold, not only titles nearest to a centroid. Appendix B gives the modeling details.

The reranker scores a slate S with member satisfaction $sat(u, i)$, defined as a within-member percentile of predicted taste fit rather than a raw rating. This avoids treating universally popular movies as automatically fair. The reward is

$$R(S) = w_1 \text{mean}_{u,i \in S} sat(u, i) + w_2 \min_u \max_{i \in S} sat(u, i) + w_3 \text{diversity}(S) - w_4 \text{var}_u \max_{i \in S} sat(u, i). \quad (2)$$

The w_2 term is the anti-veto mechanism: every member should have at least one strong option in the final slate. The greedy reranker maximizes marginal reward, while the RL reranker sequentially selects movies with a REINFORCE policy and value baseline. For a fair ablation, NN and diffusion use the identical reranker and filters; greedy and RL consume the identical candidate pools.

Mode 2 constraints such as provider, runtime, and age suitability are hard filters before scoring, not learned signals. Mode 3 cold-start profiles map declared preferences to proxy MovieLens users and are clearly treated as illustrative, not as measured scientific evidence.

Table 1. Phase2 results on 101 divergent-taste groups, $K = 5$. Higher is better except fairness gap.

Method	Rel.	Min	Gap	NDCG	Hit
Avg. baseline	0.998	1.000	0.000	0.048	0.175
NN + greedy	0.998	0.999	0.001	0.029	0.129
Diff. + greedy	0.984	0.995	0.005	0.086	0.247
NN + RL	0.987	1.000	0.000	0.072	0.243
Diff. + RL	0.887	0.997	0.003	0.133	0.391

3 Evaluation protocol

The phase2 experiment uses MovieLens 25M, with 20% of each user’s ratings held out before embedding training. These hidden ratings provide the ground truth for group NDCG@5 and Hit@5. We report divergent-taste groups as the headline because similar-taste groups are easy for nearly any method and do not stress fairness. The measured methods are: average baseline, NN plus greedy fairness reranker, diffusion plus greedy fairness reranker, NN plus RL slate-builder, and diffusion plus RL slate-builder.

This protocol avoids the circularity risk common in demos: the system does not invent a preference profile and then reward itself for matching it. Individual ratings are real; only the grouping of users into families is synthetic. The assumptions and risks are summarized in Section 6, with more detail in Appendix D.

4 Results and interpretation

Table 1 shows the main phase2 result. Diffusion plus RL delivers the best non-circular held-out quality: NDCG@5 improves from 0.048 for the average baseline to 0.133, and Hit@5 improves from 0.175 to 0.391. Against the strongest NN/RL counterpart, diffusion+RL improves NDCG@5 from 0.072 to 0.133 and Hit@5 from 0.243 to 0.391. With the greedy reranker held fixed, diffusion also improves over NN on held-out quality: NDCG@5 0.086 versus 0.029 and Hit@5 0.247 versus 0.129.

The fairness metrics require careful reading. On phase2, min-member satisfaction saturates near 1.0 for almost all methods because the MovieLens 25M catalog and percentile-based proxy provide many high-fit options. Therefore the most meaningful phase2 separation is not a dramatic min-member lift; it is that diffusion+RL substantially improves held-out retrieval quality while still keeping worst-member satisfaction at 0.997 and fairness gap at 0.003. The w_2 sweep in Appendix C confirms that increasing the worst-member weight monotonically reduces the already small fairness gap with negligible relevance loss.

Figure 1 gives the central phase2 visual read-out. It shows why the result is a compromise rather than a generic accuracy win: diffusion+RL spends some mean relevance, but it keeps worst-off satisfaction nearly saturated while delivering the best real held-out NDCG@5 and Hit@5 in Table 1. The supporting figures in Appendix C show the same story from other angles: diffusion improves held-out candidate quality under the same greedy reranker, worst-off satisfaction remains high across random/similar/divergent groups, and the fairness knob reduces the already small gap without collapsing relevance.

The project also produced a deployed app. The first page explains the model card and core modes: Mode 1 compares measured recommendation strategies, Mode 2 applies real-world filters, and Mode 3 demonstrates cold-start group building. The live app is available at <http://34.14.210.127/>.

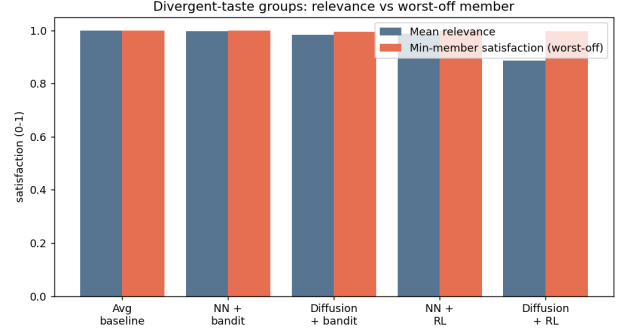


Figure 1. Phase2 divergent groups: relevance versus worst-off member satisfaction.

Table 2. How the first three pages map to supporting details.

Main-text claim	Supporting section
Pipeline and reproducibility	Appendix A
Modeling choices	Appendix B
Phase2 aggregates and figures	Appendix C
Assumptions and demo risks	Appendix D
Project and app links	Appendix E

5 System realization and demo value

The implemented system matters because the report claims are not detached from the software. The same artifacts used by evaluation are loaded by the FastAPI DemoEngine: MF embeddings, text embeddings, catalog metadata, train ratings, diffusion weights, RL policy weights, and group definitions. The React frontend is therefore a demo over cached research outputs rather than a separate hand-tuned recommendation script. This design keeps the public experience aligned with the measured pipeline.

The user-facing modes are also deliberately separated by evidential strength. Mode 1 is closest to the scientific experiment: it exposes measured groups and compares baseline, NN, diffusion, greedy, and RL strategies. Mode 2 turns the recommender into a practical movie-night tool by treating provider, runtime, and age restrictions as constraints before reranking. Mode 3 supports cold-start exploration by mapping declared group profiles to proxy MovieLens users; it is useful for explaining the idea, but it is not presented as held-out evidence because new declared profiles have no hidden ratings. This distinction is important for evaluator trust: the demo is richer than the experiment, but the paper only claims measured improvements where the data support them.

From an engineering perspective, the most useful decision was to keep the pipeline modular. Candidate generation, filtering, reward computation, greedy reranking, RL reranking, evaluation, and API serialization are separate components. That made the central ablations practical: NN and diffusion could be swapped without changing the reranker, and greedy versus RL could be compared without changing candidate generation. It also made risk handling more explicit. If TMDb enrichment is unavailable, deterministic metadata fallbacks keep the system runnable; if RL does not beat the greedy validation reward, the result is reported rather than hidden; if a group is cold-start, the UI labels it as illustrative.

6 Challenges, assumptions, and risks

The most important limitation is group ground truth. MovieLens contains individual ratings, not real household co-viewing sessions, so WATCHWISE synthesizes groups from real users. This is a standard

compromise for group recommendation research, but it means the project evaluates whether all members' hidden favorites are recovered, not whether an actual family watched together. A second limitation is the satisfaction proxy: within-member predicted percentiles are useful for fairness-aware reranking, but held-out NDCG and Hit are the stronger evidence. Third, metadata such as runtime, certification, and OTT availability may come from deterministic offline fallbacks when TMDb credentials are absent; these fields support demo filters rather than the core measured claims. Finally, RL is not unconditionally better than greedy. In phase2, RL with diffusion is best for held-out quality, but the pipeline records when the greedy validation reward is not beaten and flags the PPO contingency.

The phase2 plots sharpen these caveats instead of hiding them. The candidate-generation chart shows diffusion is valuable for finding held-out favorites, not for maximizing every proxy score. The group-difficulty chart shows why divergent groups are the meaningful test case: random and similar groups already sit near the min-member ceiling, so they cannot demonstrate much fairness lift. The candidate-space plot then gives the qualitative explanation for the quantitative result: diffusion samples a broader compromise region before snapping back to real catalog movies, whereas NN retrieval is more concentrated around the nearest centroid region. Together, the graphs support the project claim without overstating it.

7 Conclusion

WATCHWISE reframes movie recommendation from individual personalization to group compromise. The implemented system connects a real ratings backbone, diffusion candidate generation, fairness-aware reranking, non-circular evaluation, and an accessible deployed demo. The main result is nuanced but useful: diffusion is not assumed superior, yet on the phase2 held-out test it recovers real user favorites more effectively than NN retrieval under controlled comparisons, especially when paired with the RL slate-builder. The fairness proxy remains protected, practical constraints are handled as filters, and the report's appendices document the supporting details needed to reproduce and audit the claims.

A System and reproducibility details

The repository is organized as a Python research pipeline with a thin React/FastAPI demo. Core recommendation logic lives in `watchwise/`; pipeline stages live in `scripts/`; the API layer is `app/api/`; and the frontend is `app/frontend/src/`. The production app calls the Python recommendation pipeline rather than reimplementing recommendation logic in the browser.

- **Data:** MovieLens ml-25m; 25M ratings, 162K users, and 62K movies in the source dataset.
- **Artifacts:** MF embeddings, diffusion model, RL policy, catalog, ratings, groups, and results are cached under `data/cache/ml-25m/` and `results/phase2/`.
- **Main pipeline:** `python scripts/run_all.py -accelerator auto`.
- **Stage scripts:** data preparation, MF training, diffusion training, group formation, RL training, and evaluation are separated as scripts 01 through 06.
- **App launch:** `./app/start.sh`, which fetches the phase2 cache when needed and starts FastAPI plus Vite.
- **Validation:** smoke tests use `python -m pytest tests/ -q`; frontend validation uses `npm run build` from `app/frontend`.

B Model details

Embeddings. Matrix factorization learns a shared latent space for users and movies. Text features use a sentence-transformer encoder when available, with a deterministic TF-IDF plus SVD fallback so the pipeline does not depend on network access.

Diffusion candidates. The conditional diffusion component is a DDPM-style MLP denoiser over movie embeddings. It uses sinusoidal time embeddings, a cosine schedule, classifier-free guidance, and DDIM-style sampling. At inference, sampled vectors are mapped back to nearest real catalog items, ensuring every recommendation is an actual movie.

Reranking. The greedy selector builds a slate by maximizing marginal reward. The RL selector is a pointer-style REINFORCE slate policy with a value baseline, dense shaping, and entropy regularization. RL train, validation, and test groups are disjoint.

Reward semantics. Relevance is mean member satisfaction over the slate; fairness is the worst member’s best slate option. These terms are intentionally distinct: optimizing the mean alone can ignore a minority taste, while optimizing worst-member coverage alone can sacrifice group utility.

C Supporting phase2 results

Table 3. Phase2 all-group aggregate over random, similar, and divergent groups.

Method	Rel.	Min	Gap	NDCG	Hit
Avg. baseline	0.999	1.000	0.000	0.046	0.187
NN + greedy	0.998	1.000	0.000	0.030	0.119
Diff. + greedy	0.990	0.997	0.002	0.080	0.246
NN + RL	0.994	1.000	0.000	0.061	0.229
Diff. + RL	0.953	0.998	0.002	0.102	0.306

These all-group averages show the same phase2 pattern as the divergent-only headline: diffusion+RL is the strongest method on held-out retrieval quality, with NDCG@5 0.102 and Hit@5 0.306 across all 303 groups. The average baseline remains highly relevant

because many MovieLens titles are broadly liked, but its held-out retrieval quality is lower than the diffusion-based methods.

Table 4. Fairness-weight sweep for diffusion plus greedy reranking.

w_2	Relevance	Min-member	Fairness gap
0.00	0.9855	0.9945	0.0048
0.25	0.9855	0.9950	0.0043
0.50	0.9854	0.9953	0.0040
0.75	0.9854	0.9955	0.0039
1.00	0.9853	0.9956	0.0038
1.50	0.9852	0.9958	0.0035
2.00	0.9850	0.9960	0.0034

The sweep confirms that w_2 behaves like an interpretable fairness knob. Raising it from 0 to 2 moves min-member satisfaction from 0.9945 to 0.9960 and cuts the fairness gap from 0.0048 to 0.0034 while relevance changes only from 0.9855 to 0.9850.

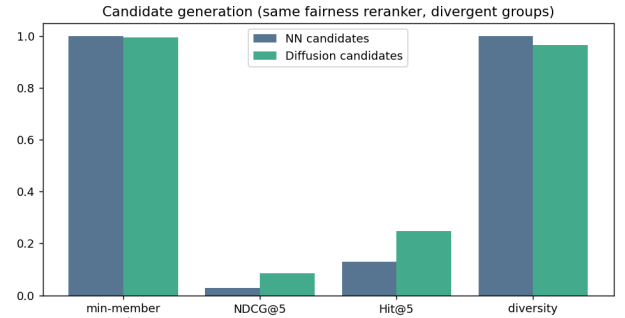


Figure 2. Phase2 candidate-generation comparison under the same greedy fairness reranker. Diffusion gives higher held-out NDCG@5 and Hit@5 than NN while keeping min-member satisfaction close to the ceiling.

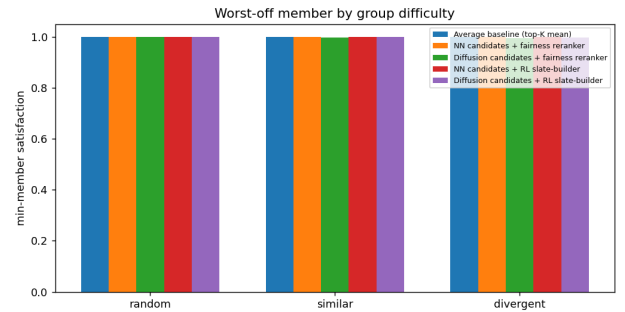


Figure 3. Phase2 worst-off satisfaction by group type. Random and similar groups are nearly solved by all methods; divergent groups are the meaningful stress test.

D Demo modes and risk handling

Mode 1: fairness stress test. This mode compares baseline, NN, diffusion, greedy, and RL methods on measured groups and exposes the model-card framing used in the project homepage.

Mode 2: OTT constraints. Provider, runtime, and age constraints are hard filters applied before scoring. This prevents the system from recommending an unavailable or inappropriate title merely because it has high predicted taste fit.

Mode 3: cold-start builder. New declared profiles are mapped to proxy MovieLens users. This is useful for demonstration but not

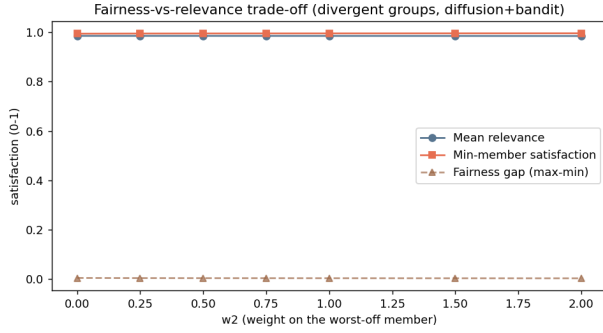


Figure 4. Increasing the fairness weight reduces the already small fairness gap with minimal relevance movement.

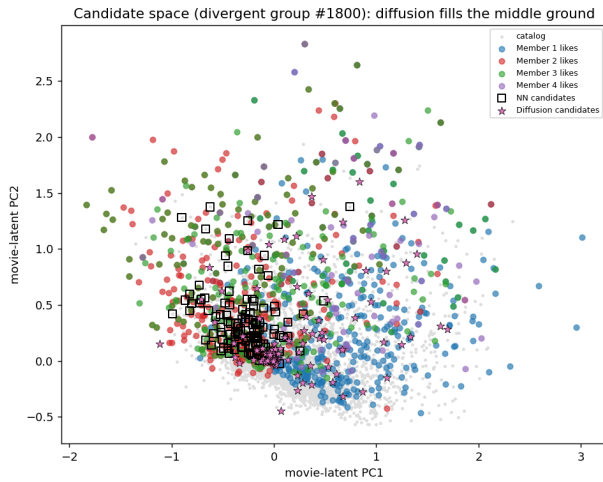


Figure 5. Phase2 candidate-space projection for a divergent group. NN candidates cluster near one retrieval region, while diffusion samples spread into the middle-ground candidate space before mapping back to real movies.

counted as held-out evaluation because new profiles have no hidden real ratings.

Risk controls. The project keeps generated caches out of source control, labels deterministic metadata fallbacks, preserves disjoint group splits for RL, and keeps the deployed app as a consumer of cached pipeline artifacts rather than a separate recommendation implementation.

E Project links

Project homepage: <https://github.com/WatchWise-IISc/WatchWise>

Live application: <http://34.14.210.127/>